

Anak Agung Krisna Ananda Kusuma (Krisna)

Mobile: +6281319771165 | Email: krisnakusuma0500@gmail.com |

Highly self-motivated fresh graduate with over five years of experience in robotics development pipeline, including perception, planning, and control. Specialized in model-based and model-free control systems. Skilled in robotic simulation, hardware interfaces, and real-world deployment. Aiming to hone expertise in cutting-edge robotics applications by working alongside experienced industry experts to deliver research-driven development as real-world solutions.

EDUCATION

Master of Science, Seoul National University of Science and Technology **2023 – 2026**

Thesis title : Robust Locomotion Control for Quadruped Robot using Deep Learning

GPA : 4.31/4.50

Double Degree Program, Universität Duisburg-Essen, Germany **2021 – 2023**

Thesis title : Safe Navigation of a Differential Wheeled Robot using Model Predictive Control for Local Trajectory Planning and Collision Avoidance

Grade : 1.7/1.0

Bachelor of Science, Universitas Indonesia **2018 – 2023**

GPA : 3.72/4.0

WORKING EXPERIENCE

Seoul National University of Science and Technology **Sept 2023 – Feb 2026**

Research Assistant (Control and Intelligent Robotics Lab)

- Conducted research on robust imitation learning of quadruped robot locomotion control derived from an MPC expert, implemented in Gazebo simulation using C++ and Python.
- Conducted research about mapless navigation and control for autonomous racecar, implemented in Gazebo simulation using Python, tested on real world hardware.
- Develop the perception, planning, and control pipeline for the F1Tenth platform in both Gazebo simulation and real-world scenarios, followed by competing in the F1Tenth Korea championship.
- Mentored students, delivered learning materials, and provide homework assignments and solutions for undergraduate courses in machine learning and robotics class.

Published first authored papers:

- Receding Horizon Smooth Trajectory Generation and Tracking for Autonomous Mapless Racing – <https://doi.org/10.1007/s12555-025-0581-9>
- Implementation of Domain Randomization for Quadrupedal Robot Locomotion – <https://www.dbpia.co.kr/pdf/pdfView.do?nodeId=NODE12051710>

Universität Duisburg-Essen **Oct 2022 – Mar 2023**

Student Research Assistant, (6-months contract)

- Implement software packages in Python and ROS for a PhD student's research on robot navigation in crowded environments using Zed 2 camera and LoCoBot robot.
- Support students in labs, projects, and theses by providing assistance in ROS.
- Design and print 3D-components for robots using Autodesk Inventor.

Telkom Indonesia**Jan 2021 – Mar 2021***LoRa Networks Monitoring Intern*

- Developed an automated system for hourly status reporting of a LoRa network operated by Telkom Indonesia using Python.

IcanStudioLive Concert Studio**Jul 2020 – Aug 2020***Electrical Control Engineer (Part-time)*

- Designed and fabricate two electrical system for the control of two ceiling dolly using Arduino and NRF24L01 wireless transceiver module.

Faculty of Engineering Universitas Indonesia**Jun 2020 – Jul 2020***PCB Designer (Part-time)*

- Designed a PCB for the first prototype of a Covent-20 ventilator developed by Faculty of Engineering Universitas Indonesia using Altium designer.

ORGANIZATIONAL EXPERIENCE

Tim Robotika Universitas Indonesia (TRUI), Universitas Indonesia**Jan 2021 – Oct 2021***President*

- Led the management of twelve teams within TRUI, including annual proposals and budget planning, social media documentation, oversight of yearly final reports, and communication with the Directorate of Student Affairs at Universitas Indonesia.
- Organized open recruitment for TRUI and held electrical, mechanical, and programming training.
- Organized leadership training for prospective TRUI presidents by inviting TRUI alumni, the supervising professor, and former TRUI presidents.

Gladiatos, Humanoid Robot Team Universitas Indonesia**Nov 2020 – Oct 2021***Leader & Head of Technical Division*

- Led the development of two humanoid robot using Robotis OP2 and coordinated task assignments across electrical, mechanical, programming, and management divisions.
- Managed weekly cross-division meetings, tracked progress, created annual budget and development plans, and provide regular updates to the supervising professor.
- Conducted introductory and training sessions on humanoid robots for new members joining the Gladiatos team.

HONORS AND AWARDS

- Huawei Seeds For The Future program awardee | 2022
- FGA Digital Talent Scholarship awardee from Kominfo Indonesia | 2021
- Roberto Rocca Education Program scholarship awardee from Tenaris | 2021
- Winner of Lomba Karya Cipta Teknologi (LKCT) ELCCO held by Universitas Udayana | 2020

SKILLS AND LANGUAGES

Hard Skills	: ROS, Linux ubuntu, Python, C/C++, Gazebo, Isaac Sim, Pytorch, Arduino.
Soft Skills	: Leadership, Problem Solving, Critical Thinking, Teamwork, Initiative, Adaptable.
Languages	: English (Proficient User), German (Beginner), Bahasa Indonesia (Native)
Certifications	: Huawei HCIA AI certification, IELTS Overall Band Score 7.0/9.0
Specializations	: Control systems, simulation-to-real deployment, AI/ML for robotics.